MATHEMATICS EXCELLENCE ASSOCIATION OF TEACHERS

(WE'RE UNLEASHING MATHEMATICS POTENTIAL)

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A'LEVEL PRINCIPAL MATHEMATICS SYLLABUS

Senior 5 Principal maths Syllabus

Term	Topic		Sub-topics		Periods
Term 1	1.	Numerical concepts	1.1	Indices	10
			1.2	Logarithms	
			1.3	Surds	
	2.	Equations and inequalities	2.1	Linear and simultaneous equations	26
			2.2	Quadratic equations	
			2.3	Quadratic inequalities	
			2.4	Polynomials	
	3.	Coordinate geometry	3.1	Straight lines	12
	4.	Partial fractions	4.1	Linear factor	08
			4.2	Quadratic factor	
			4.3	Repeated factor	
			4.4	Improper fraction	
	5.	Trigonometry	5.1	Trigonometrical ratios	25
			5.2	Clockwise and anti-clockwise angles	
			5.3	Graphs of $\sin\theta$, $\cos\theta$, $\tan\theta$	
			5.4	Compound angle formulae and derived identities	
			5.5	Solution of triangles	
	6.	Descriptive statistics	6.1	Measures of dispersion	15
			6.2	Measures of relative positions	
	Total				96

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Term	Topic		Sub-topics		Periods
Term 2	7.	Correlation and scatter diagrams	7.1	Scatter diagram	08
			7.2	Correlation	
		8. Dynamics 1	8.1	Resultant and components of forces	20
			8.2	Friction	
			8.3	Connected particles	
	9.	Probability theory	9.1	Probability theorems	16
			9.2	Applications of probability in real life	
	10.	Differentiation 1	10.1	Gradient of a curve	20
			10.2	Gradient functions	
			10.3	Composite functions	
			10.4	Implicit functions	
			10.5	Products and quotients of functions	
			10.6	Applications of differentiation1 (Nature of turning points, curve sketching, small changes, rates of change and velocity and acceleration)	
	11.	Integration 1	11.1	Indefinite integral	22
			11.2	Definite integral	
			11.3	Applications of integration (Area enclosed in stated boundaries and volume of solids of revolution)	
			11.4	Mean value of a function	
	12.	Permutations and combinations	12.1	Permutations and combinations	10
	Total				96

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Term	Topic		Sub-topics		Periods
Term 3	13.	Series	13.1	Arithmetic Progression (A.P)	20
			13.2	Geometric Progression (G.P)	
			13.3	Proof by induction	
			13.4	Binomial expansion	
	14.	Random variables	14.1	Discrete random variables	30
			14.2	Continuous random variables	
		15. Probability distributions	15.1	Binomial distribution	30
			15.2	Uniform rectangular distribution.	
			15.3	Normal distribution	
			15.4	Normal approximation to the binomial distribution for n>20 30	
	16.	Error analysis		16.1 Errors	16
				16.2 Propagation of errors using simple interval arithmetic method	
				16.3 Errors in functions	
	Total				96

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Term	Topic		Sub-topics		Periods
Term 1	17.	Vectors	17.1	Vectors in three dimensions	22
			17.2	Lines in two and three dimensions	
			17.3	Planes	
	18.	Differentiation 2	18.1	Trigonometric functions	40
			18.2	Exponential, logarithmic and inverse trigonometric functions	
			18.3	Maclaurin's theory	
			18.4	Further curve sketching	
	19.	Integration 2	19.1	Function and its derivative (change of variables)	34
			19.2	Exponential and logarithmic functions	
			19.3	Trigonometric functions (Even and odd powers of $\sin\theta$, $\cos\theta$, etc, Factor formulae, Inverse trigonometric functions and t-substitution)	
			19.4	Partial fractions	
				19.5 Integration by parts	
	Total				96

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Term	Topic		Sub-topics		Periods
Term 2	20.	Dynamics 2	20.1	Resultant velocity	28
			20.2	Relative motion	
			20.3	Projectiles	
	21.	Trapezium rule	21.1	Estimating an Integral	06
			21.2	Percentage error	
	22.	Sampling distribution	22.1	Distribution of sampling mean	12
			22.2	Point estimation	
			22.3	Interval estimation	
	23.	Iterative methods	23.1	Interpolation and extrapolation	22
			23.2	Location of roots	
			23.3	Newton Raphson method	
			23.4	Further linear interpolation	
	24.	Coordinate geometry 2	24.1	Locus	28
			24.2	The circle	
			24.3	Parabola	
			24.4	Ellipse	
	Total				96

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Term 3	25.	Complex numbers	25.1	Imaginary numbers	28
			25.2	Algebra of complex numbers	
			25.3	Argand diagram and polar form	
			25.4	Locus	
			25.5	De Moivre' s theorem	
	26.	Differential equations	26.1	Differential equations	26
			26.2	Solving first order differential equations	
			26.3	Application of differential equations	
	27.	Flow charts	27.1	Algorithms and presenting them on flow charts	16
			27.2	Performing dry runs	
	Total				70

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